

applied to different environments, to achieve operational compliance rather than perfection. Choose the right SLAM algorithm for your purpose depending on the kind of observation to be made, that is, sensor data and map type required. The sensor could be anything, like a 2D laser scanner or a range-only sensor, while mapping types range across occupancy grids, point maps, landmark maps and even graphical representations.

Algorithms aplenty. You have Bayesian range-only SLAM, iterative closest point (ICP) SLAM, particle filters like Rao-Blackwellised particle filtering (RBPF) SLAM, visual SLAM based on maximum likelihood estimation like Bundle-Adjustment, extended Kalman filter, so on and so forth; mix and match to get the right results.

SLAM being a popular research area, published algorithms find

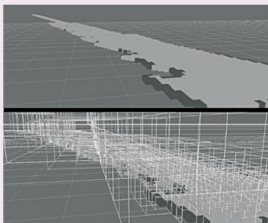


Fig. 3: COctoMapVoxels on MRPT

implementations in self-driving cars, drones, autonomous underwater vehicles, planetary rovers, domestic robots and even inside the human body.

Make your own robot

Common robotic platforms and sensors, Monte Carlo localisation, reac-

tive navigation and such details have their own off-the-shelf modules that come as part of this open source suite, forming the perfect base for beginning a new robotic application from scratch. The toolkit also comes with Robot operating system (ROS) wrapper packages, coming as a boon to designers working on ROS. Before you begin, have a look at the list of sensors and platforms supported by the toolkit, on their website. The only thing that now limits your robot becoming a super robot is the code. **EFY**

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